

Explainable Clinical Risk Prediction Using Machine Learning and SHAP

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Healthcare AI | Explainable AI | Machine Learning

1. Introduction

Heart disease remains one of the leading causes of global mortality and early prediction plays a critical role in preventive healthcare and clinical decision-making. Artificial Intelligence (AI) and Machine Learning (ML) methods have increasingly been adopted in healthcare to improve disease prediction accuracy. However, many predictive models behave as “black-box” systems, limiting transparency and clinician trust.

This project explores interpretable machine learning approaches for heart disease prediction using the UCI Heart Disease dataset. In addition to predictive modeling, Explainable AI (XAI) techniques using SHAP (SHapley Additive exPlanations) were implemented to provide transparent explanations of model predictions.

The project compares Logistic Regression and Random Forest classifiers while evaluating interpretability, prediction performance, and feature influence.

2. Dataset Description

The UCI Heart Disease dataset containing 1025 patient records and 14 clinical attributes was used for this project.

Dataset Shape

- Rows: 1025
- Columns: 14

Target Variable

- 1 → Presence of heart disease
- 0 → Absence of heart disease

Target Distribution

- Positive cases: 526
- Negative cases: 499

The dataset was relatively balanced, reducing the risk of severe class imbalance during model training.

Important Features

- Age
- Sex
- Chest pain type (cp)
- Resting blood pressure (trestbps)
- Cholesterol (chol)
- Maximum heart rate achieved (thalach)
- Exercise-induced angina (exang)
- ST depression (oldpeak)
- Number of major vessels (ca)

Data Quality

Missing value analysis showed:

- No missing values were present in the dataset.

Statistical Summary

The dataset showed:

- Mean patient age: 54.43 years
- Mean cholesterol level: 246
- Mean resting blood pressure: 131.61

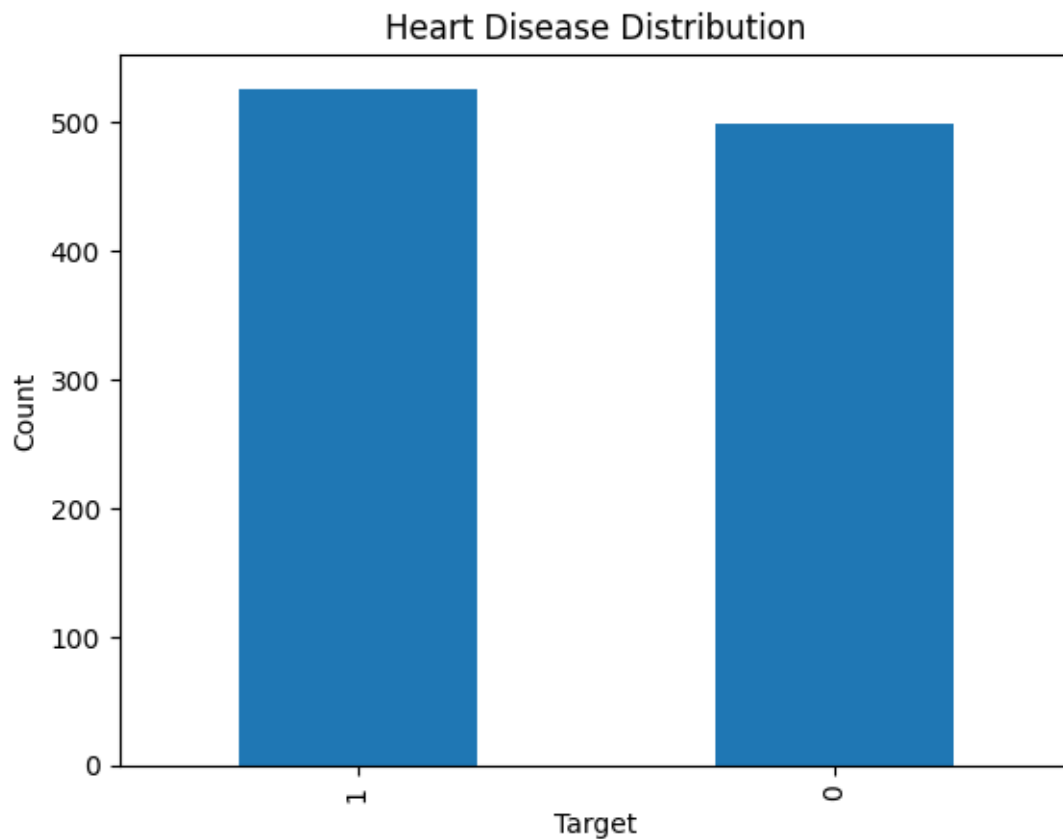
These values suggest the dataset contains clinically relevant cardiovascular risk patterns.

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand feature distributions and relationships between clinical variables.

Target Distribution Analysis

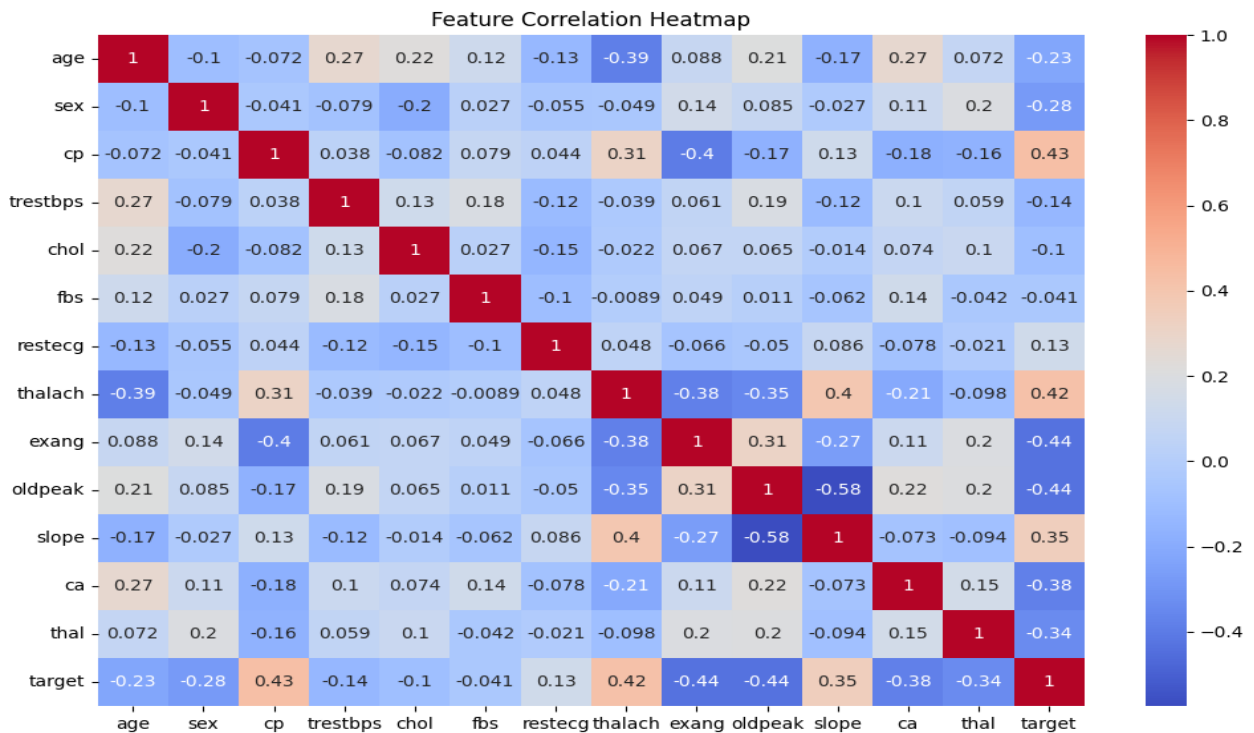
The dataset showed nearly balanced disease and non-disease cases.



This balance helps improve model stability and reduces bias toward a dominant class.

Correlation Heatmap Analysis

A correlation heatmap was generated to identify relationships between variables.



The heatmap revealed meaningful associations between:

- chest pain type (cp)
- maximum heart rate (thalach)
- ST depression (oldpeak)
- exercise-induced angina (exang)

and heart disease prediction outcomes.

Positive correlations indicated variables that increased with heart disease occurrence, while negative correlations suggested inverse relationships.

The heatmap also helped identify clinically influential variables prior to model training.

4. Machine Learning Methodology

The dataset was divided into:

- Training Set: 820 samples
- Testing Set: 205 samples

Two machine learning models were implemented:

Logistic Regression

Logistic Regression was used as an interpretable baseline model. Feature scaling was applied using StandardScaler before training.

Logistic Regression Results

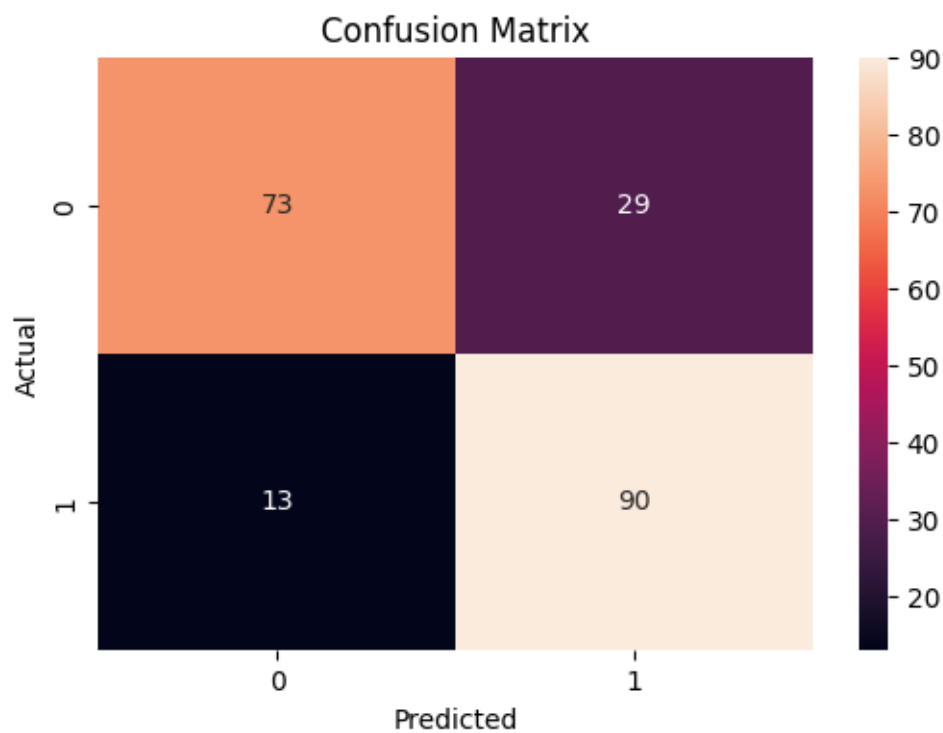
- Accuracy: 79.51%
- ROC-AUC Score: 0.8787

Classification Performance

- Precision (Disease): 0.76
- Recall (Disease): 0.87
- F1-score (Disease): 0.81

These results indicate that the model successfully identified most disease-positive patients while maintaining acceptable precision.

Confusion Matrix Analysis

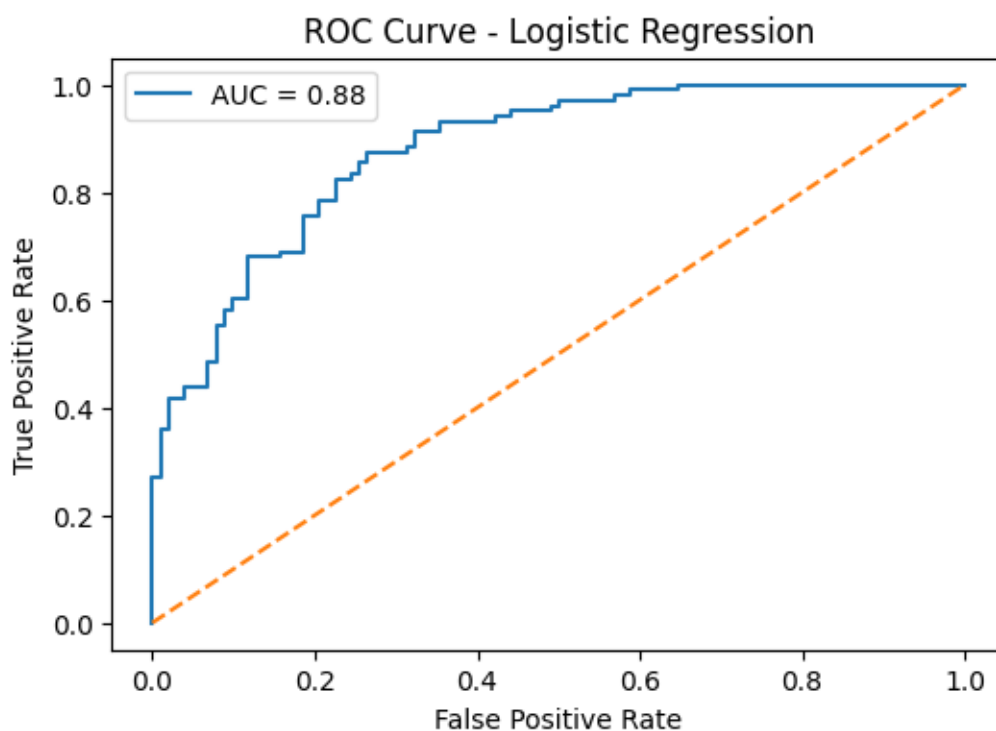


The confusion matrix demonstrated:

- strong true positive detection,
- moderate false positives,
- relatively low false negatives.

In healthcare AI systems, minimizing false negatives is especially important because missed disease predictions may delay diagnosis and treatment.

ROC Curve Analysis



The ROC curve demonstrated good discrimination capability between healthy and diseased patient classes.

An AUC score of 0.8787 indicates strong predictive performance beyond random classification.

5. Random Forest Model

Random Forest was implemented to improve nonlinear pattern recognition and prediction performance.

Random Forest Results

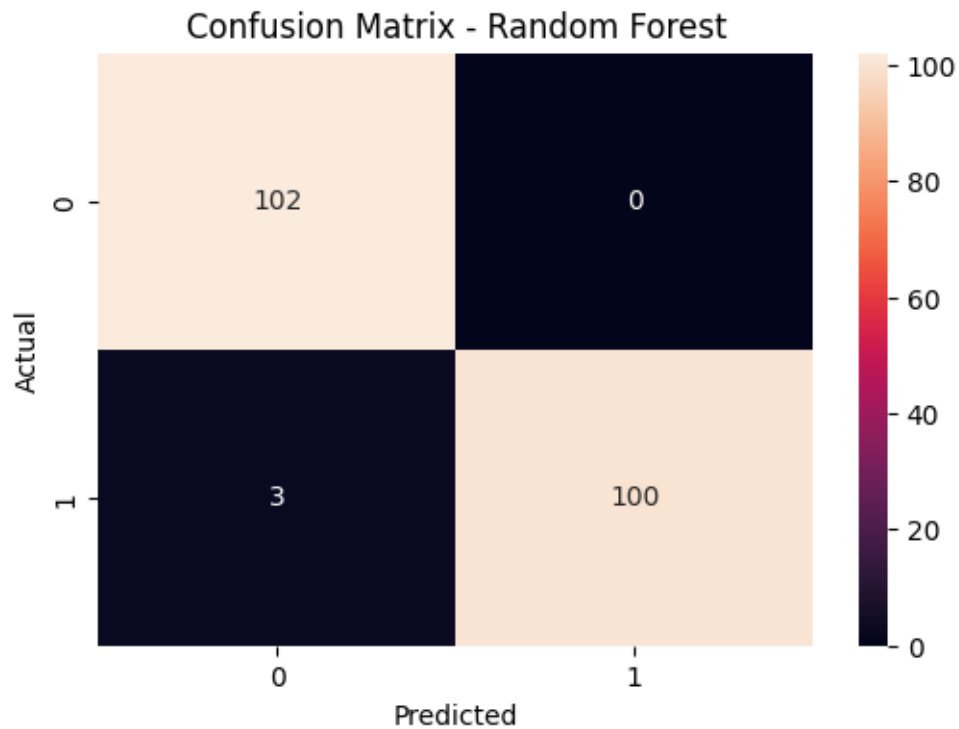
- Accuracy: 98.54%
- ROC-AUC Score: 1.00

Classification Performance

- Precision (Disease): 1.00
- Recall (Disease): 0.97
- F1-score (Disease): 0.99

The Random Forest model significantly outperformed Logistic Regression by capturing complex feature interactions within the dataset.

Random Forest Confusion Matrix

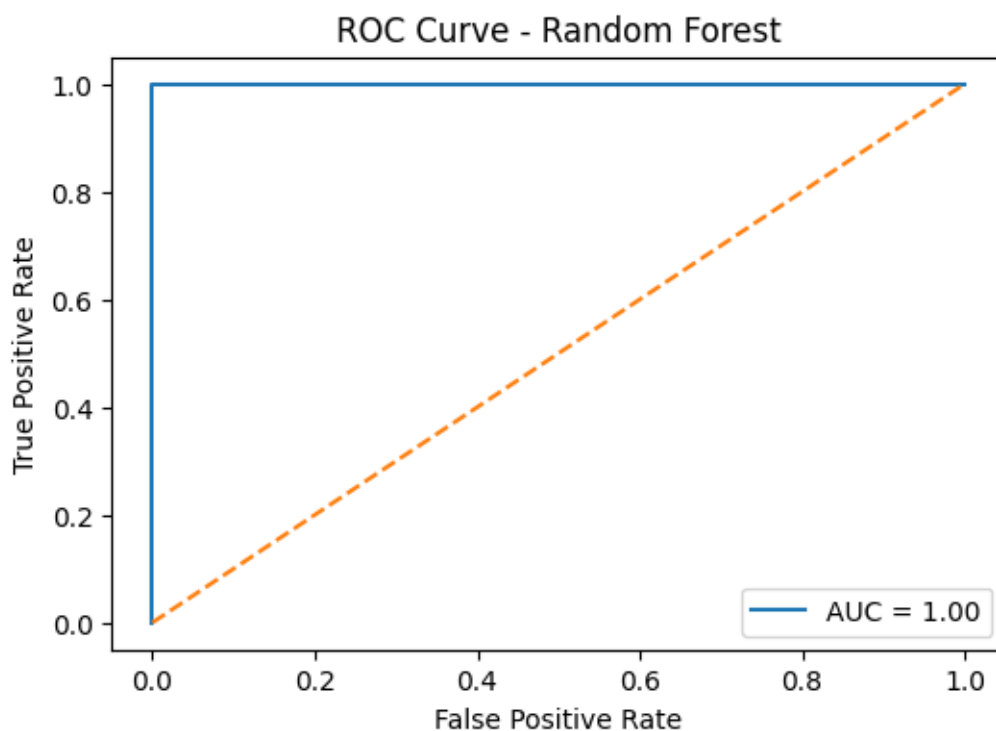


The Random Forest confusion matrix showed:

- extremely high true positive rates,
- near-perfect classification,
- minimal false negatives.

This suggests strong predictive capability for cardiovascular risk classification.

Random Forest ROC Curve



The ROC-AUC score of 1.00 indicates near-perfect separation between disease-positive and disease-negative classes on the testing dataset.

Although highly effective, such performance should still be interpreted cautiously due to the possibility of overfitting on benchmark datasets.

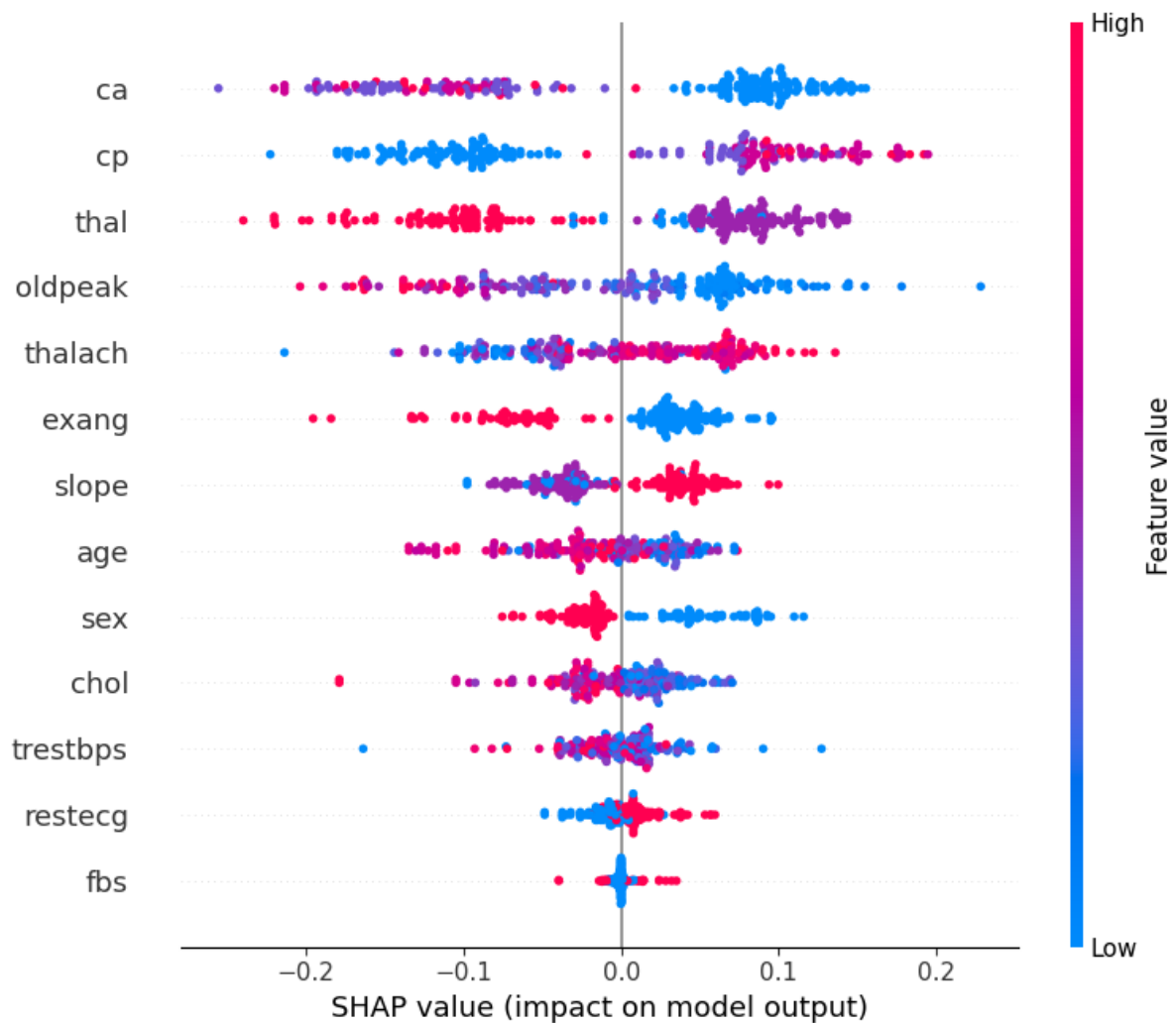
6. Explainable AI using SHAP

To improve model transparency and interpretability, SHAP explainability analysis was applied to the Random Forest model.

SHAP helps explain:

- how each feature influences predictions,
- why the model predicts disease risk,
- which variables contribute most strongly to outcomes.

SHAP Beeswarm Plot



The SHAP summary plot identified the most influential features:

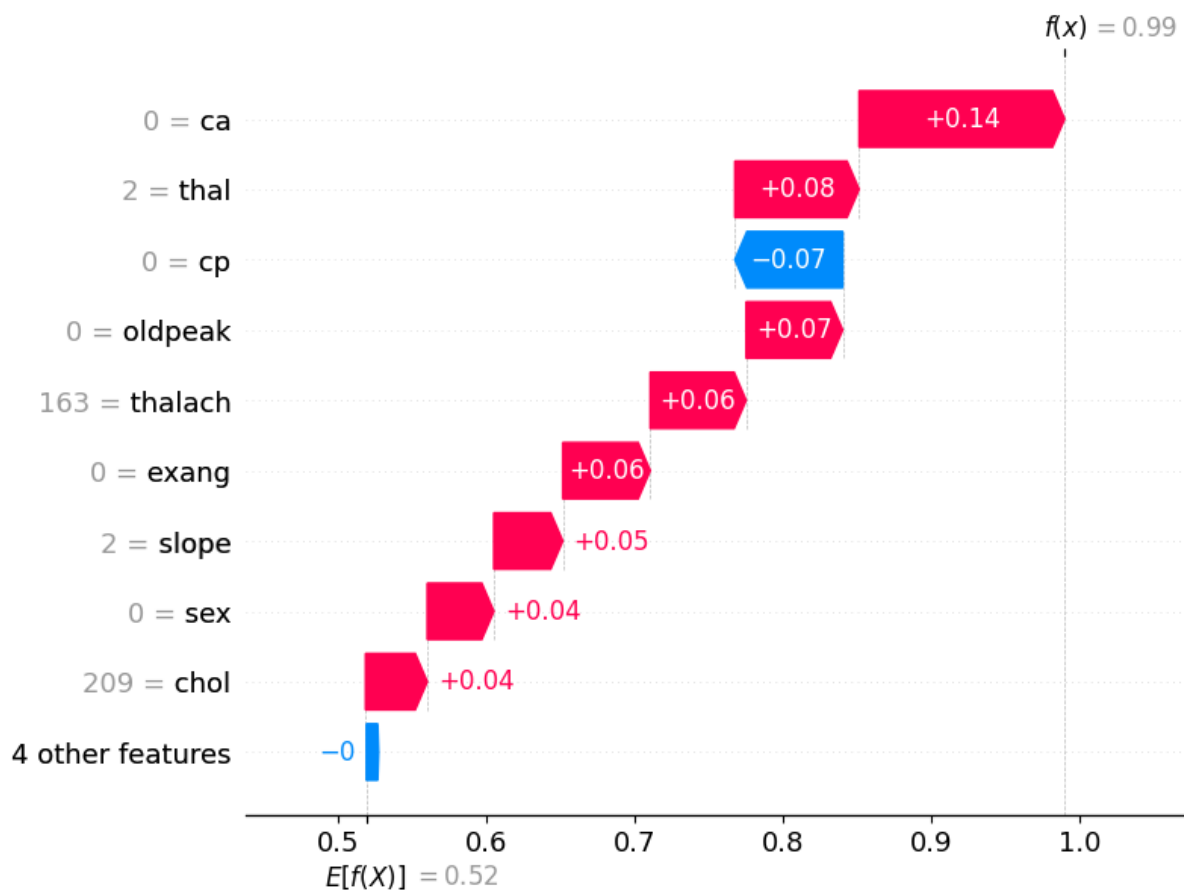
1. Chest pain type (cp)
2. Number of major vessels (ca)
3. Maximum heart rate (thalach)
4. ST depression (oldpeak)
5. Thalassemia (thal)

Healthcare Interpretation

- Higher oldpeak values strongly pushed predictions toward heart disease.
- Abnormal chest pain categories contributed significantly to positive disease predictions.
- Reduced maximum heart rate values increased cardiovascular risk predictions.

These findings align with known cardiovascular clinical risk factors.

SHAP Waterfall Plot



The waterfall plot provided patient-level explanations by showing:

- which features pushed predictions toward disease,
- which variables reduced disease probability,
- magnitude of individual feature influence.

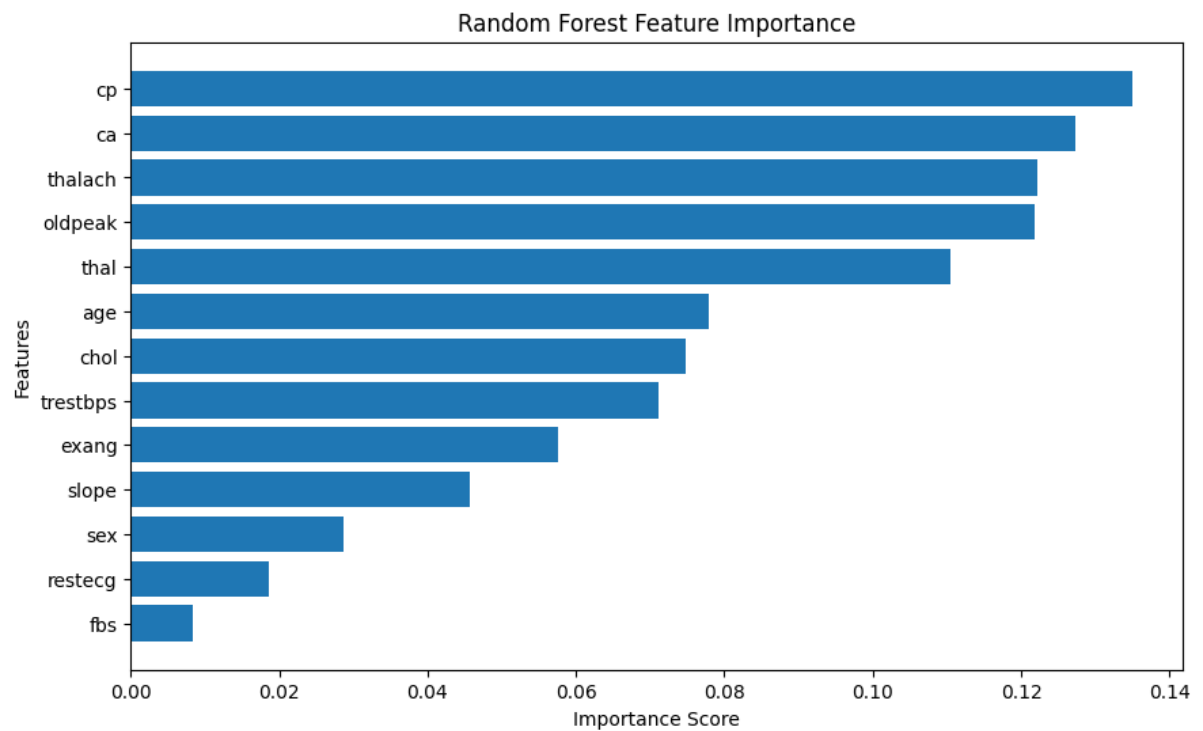
This improves trust and interpretability in healthcare AI systems.

7. Feature Importance Analysis

Feature importance analysis from the Random Forest model identified the strongest predictive variables.

Top Features

Feature	Importance
cp	0.135
ca	0.127
thalach	0.122
oldpeak	0.121
thal	0.111



The analysis showed that chest pain type and major vessel count were among the strongest contributors to prediction outcomes.

Feature importance analysis complements SHAP explainability by identifying globally influential features within the model.

8. Model Comparison

Model	Accuracy	ROC-AUC
Logistic Regression	79.51%	0.8787
Random Forest	98.54%	1.0000

The Random Forest model demonstrated significantly higher predictive performance than Logistic Regression due to its ability to capture nonlinear relationships and feature interactions.

However, explainability remains critical in healthcare AI systems, making SHAP-based interpretation highly valuable.

9. Limitations

Several limitations should be considered:

- The dataset originated from a public benchmark dataset and may not fully represent real-world hospital populations.
- External clinical validation was not performed.
- Real-time deployment and calibration analysis were outside the scope of this project.
- High Random Forest performance may indicate potential overfitting to benchmark data.

10. Future Work

Future improvements may include:

- external hospital validation,
- multimodal healthcare data integration,

- calibration analysis,
- fairness and bias evaluation,
- deep learning approaches,
- longitudinal patient monitoring systems.

Further research may also explore clinician-in-the-loop explainable AI systems for real-world healthcare applications.

11. Conclusion

This project demonstrated the application of interpretable machine learning techniques for cardiovascular risk prediction.

Logistic Regression provided transparent baseline predictions, while Random Forest significantly improved predictive performance through nonlinear ensemble learning. SHAP explainability techniques enabled transparent interpretation of model decisions, supporting trust and interpretability in healthcare AI applications.

The project highlights the importance of combining predictive performance with explainability for future trustworthy clinical decision-support systems.